

OUTLINE

1

**INTRODUCTION TO BIAS AND
FAIRNESS IN AI**



2

FAIRNESS MEASURES



3

BIAS MITIGATION



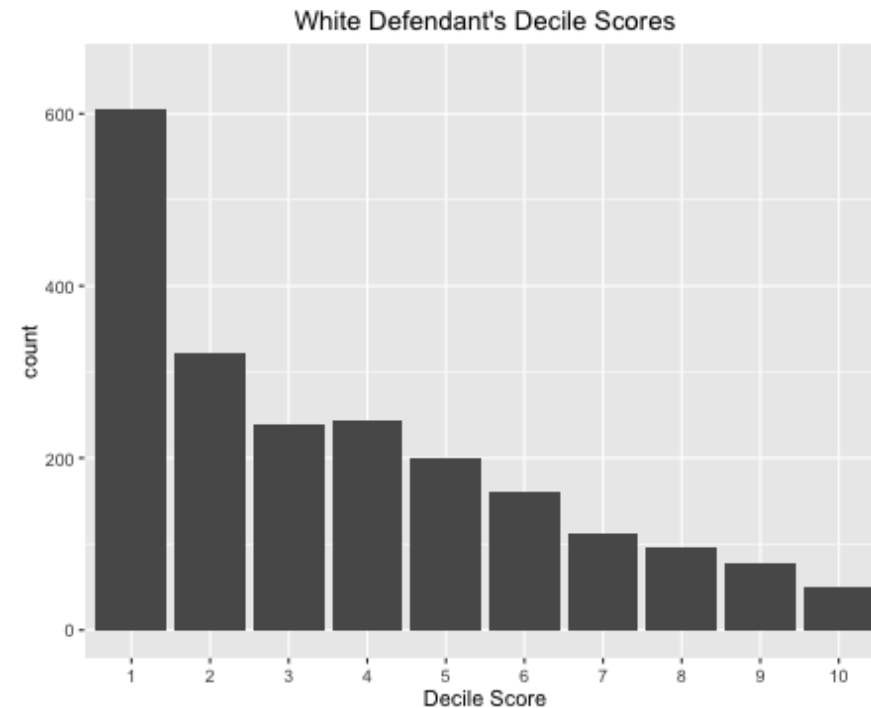
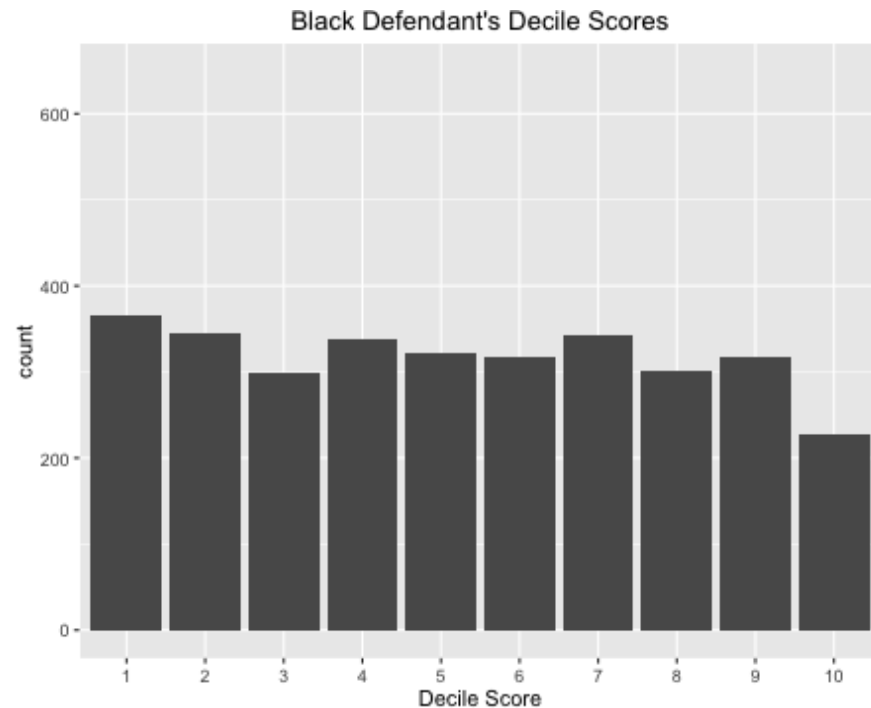
CASE STUDY: COMPAS

1. PERSON COMMITS A CRIME, IS ARRESTED
2. COMPAS SOFTWARE PREDICTS THE CHANCE THAT THE PERSON WILL COMMIT ANOTHER CRIME IN THE FUTURE (RECIDIVISM)
3. RECIDIVISM SCORES IMPACT CRIMINAL SENTENCES: IF A PERSON IS LIKELY TO COMMIT ANOTHER CRIME, SHOULDN'T THEY GET A LONGER SENTENCE?

CASE STUDY: COMPAS

“Our analysis of Northpointe’s tool, called COMPAS (which stands for Correctional Offender Management Profiling for Alternative Sanctions), found that black defendants were far more likely than white defendants to be incorrectly judged to be at a higher risk of recidivism, while white defendants were more likely than black defendants to be incorrectly flagged as low risk”

CASE STUDY: COMPAS



Question: How many of these people ended up committing new crimes within 2 years?

Source: <https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm>

ERROR METRICS

	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVIDSM	TRUE NEGATIVE (TN)	FALSE POSITIVE (FP)
OUTCOME: RECIDIVATED	FALSE NEGATIVE (FN)	TRUE POSITIVE (TP)

ERROR RATE

	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVDSM	TRUE NEGATIVE (TN)	FALSE POSITIVE (FP)
OUTCOME: RECIDIVATED	FALSE NEGATIVE (FN)	TRUE POSITIVE (TP)

$$\text{ERROR RATE} = \frac{FP + FN}{TN + FP + FN + TP}$$

How often is the prediction wrong?

FALSE POSITIVE RATE

	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVIDSM	TRUE NEGATIVE (TN)	FALSE POSITIVE (FP)
OUTCOME: RECIDIVATED	FALSE NEGATIVE (FN)	TRUE POSITIVE (TP)

$$\text{ERROR RATE} = \frac{FP + FN}{TN + FP + FN + TP}$$

How often is the prediction wrong?

$$\text{FALSE POSITIVE RATE} = \frac{FP}{FP + TN}$$

How often were non-offenders predicted to reoffend?

FALSE NEGATIVE RATE

	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVDSM	TRUE NEGATIVE (TN)	FALSE POSITIVE (FP)
OUTCOME: RECIDIVATED	FALSE NEGATIVE (FN)	TRUE POSITIVE (TP)

$$\text{ERROR RATE} = \frac{FP + FN}{TN + FP + FN + TP}$$

How often is the prediction wrong?

$$\text{FALSE POSITIVE RATE} = \frac{FP}{FP + TN}$$

How often were non-offenders predicted to reoffend?

$$\text{FALSE NEGATIVE RATE} = \frac{FN}{FN + TP}$$

How often were offenders predicted not to reoffend?

DIFFERENT STAKEHOLDERS

	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVDSM	TRUE NEGATIVE (TN)	FALSE POSITIVE (FP)
OUTCOME: RECIDIVATED	FALSE NEGATIVE (FN)	TRUE POSITIVE (TP)

$$\text{ERROR RATE} = \frac{FP + FN}{TN + FP + FN + TP}$$

How often is the prediction wrong?

$$\text{FALSE POSITIVE RATE} = \frac{FP}{FP + TN}$$

How often were non-offenders predicted to reoffend?

$$\text{FALSE NEGATIVE RATE} = \frac{FN}{FN + TP}$$

How often were offenders predicted not to reoffend?

DIFFERENT STAKEHOLDERS

	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIDIVISM	TRUE NEGATIVE (TN)	FALSE POSITIVE (FP)
OUTCOME: RECIDIVATED	FALSE NEGATIVE (FN)	TRUE POSITIVE (TP)

$$\text{ERROR RATE} = \frac{FP + FN}{TN + FP + FN + TP}$$

How often is the prediction wrong?

Defendants
care about this

$$\text{FALSE POSITIVE RATE} = \frac{FP}{FP + TN}$$

How often were non-offenders
predicted to reoffend?

$$\text{FALSE NEGATIVE RATE} = \frac{FN}{FN + TP}$$

How often were offenders
predicted not to reoffend?

DIFFERENT STAKEHOLDERS

	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVDSM	TRUE NEGATIVE (TN)	FALSE POSITIVE (FP)
OUTCOME: RECIDIVATED	FALSE NEGATIVE (FN)	TRUE POSITIVE (TP)

$$\text{ERROR RATE} = \frac{FP + FN}{TN + FP + FN + TP}$$

How often is the prediction wrong?

Defendants
care about this

$$\text{FALSE POSITIVE RATE} = \frac{FP}{FP + TN}$$

How often were non-offenders
predicted to reoffend?

Judges care
about this

$$\text{FALSE NEGATIVE RATE} = \frac{FN}{FN + TP}$$

How often were offenders
predicted not to reoffend?

CASE STUDY: COMPAS

	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVIDSM	2681 (TN)	1282 (FP)
OUTCOME: RECIDIVATED	1216 (FN)	2035 (TP)

$$\text{ERROR RATE} = \frac{FP + FN}{TN + FP + FN + TP} \approx 34.6\%$$

$$\text{FALSE POSITIVE RATE} = \frac{FP}{FP + TN} \approx 32.4\%$$

$$\text{FALSE NEGATIVE RATE} = \frac{FN}{FN + TP} \approx 37.4\%$$

CASE STUDY: COMPAS

BLACK DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVIDSM	990 (TN)	805 (FP)
OUTCOME: RECIVATED	532 (FN)	1369 (TP)

WHITE DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVIDSM	1139 (TN)	349 (FP)
OUTCOME: RECIVATED	461 (FN)	505 (TP)

CASE STUDY: COMPAS

BLACK DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK	WHITE DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVIDSM	990 (TN)	805 (FP)	OUTCOME: NO RECIVIDSM	1139 (TN)	349 (FP)
OUTCOME: RECIVATED	532 (FN)	1369 (TP)	OUTCOME: RECIVATED	461 (FN)	505 (TP)

$$\text{ERROR RATE} = \frac{FP + FN}{TN + FP + FN + TP}$$

$$\text{FALSE POSITIVE RATE} = \frac{FP}{FP + TN}$$

$$\text{FALSE NEGATIVE RATE} = \frac{FN}{FN + TP}$$

CASE STUDY: COMPAS

BLACK DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVIDSM	990 (TN)	805 (FP)
OUTCOME: RECIVATED	532 (FN)	1369 (TP)

ERROR RATE= $\approx 36.2\%$

WHITE DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIVIDSM	1139 (TN)	349 (FP)
OUTCOME: RECIVATED	461 (FN)	505 (TP)

$\approx 33\%$

Similar error rates between
white and black defendants

CASE STUDY: COMPAS

BLACK DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIDIVISM	990 (TN)	805 (FP)
OUTCOME: RECIDIVATED	532 (FN)	1369 (TP)

WHITE DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIDIVISM	1139 (TN)	349 (FP)
OUTCOME: RECIDIVATED	461 (FN)	505 (TP)

ERROR RATE = $\approx 36.2\%$

$\approx 33\%$

FALSE POSITIVE RATE = $\approx 44.9\%$

$\approx 23.5\%$

Black defendants have 1.9x higher False Positive Rate!

CASE STUDY: COMPAS

BLACK DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIDIVISM	990 (TN)	805 (FP)
OUTCOME: RECIDIVATED	532 (FN)	1369 (TP)

WHITE DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIDIVISM	1139 (TN)	349 (FP)
OUTCOME: RECIDIVATED	461 (FN)	505 (TP)

ERROR RATE = $\approx 36.2\%$

$\approx 33\%$

FALSE POSITIVE RATE = $\approx 44.9\%$

$\approx 23.5\%$

FALSE NEGATIVE RATE = $\approx 28\%$

$\approx 47.7\%$

White defendants have 1.7x higher False Negative Rate

CASE STUDY: COMPAS

BLACK DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIDIVISM	990 (TN)	805 (FP)
OUTCOME: RECIDIVATED	532 (FN)	1369 (TP)

WHITE DEFENDANTS	PREDICTION: LOW RISK	PREDICTION: HIGH RISK
OUTCOME: NO RECIDIVISM	1139 (TN)	349 (FP)
OUTCOME: RECIDIVATED	461 (FN)	505 (TP)

ERROR RATE = $\approx 36.2\%$

FALSE POSITIVE RATE = $\approx 44.9\%$

FALSE NEGATIVE RATE = $\approx 28\%$

$\approx 33\%$

$\approx 23.5\%$

$\approx 47.7\%$

Surprising fact: COMPAS gives very different outcomes for white vs black defendants, but it does not use ethnicity as an input to the algorithm!

**WHY CAN'T WE JUST OMIT
ANY PROTECTED
ATTRIBUTES FROM THE
DATASET?**

LATENT AND PROXY VARIABLES

KEY DEFINITIONS

LATENT VARIABLES: UNOBSERVED FACTORS THAT INFLUENCE OUTCOMES BUT AREN'T DIRECTLY MEASURED IN THE DATA. THEY CAN INTRODUCE BIAS IF NOT PROPERLY ACCOUNTED FOR

PROXY VARIABLES: OBSERVABLE ATTRIBUTES THAT, ALTHOUGH NOT DIRECTLY SENSITIVE (LIKE GENDER OR ETHNICITY), ARE HIGHLY CORRELATED WITH PROTECTED ATTRIBUTES AND CAN ACT AS THEIR SUBSTITUTES

IMPLICATIONS FOR FAIRNESS IN ML

- ❑ SIMPLY REMOVING PROTECTED ATTRIBUTES DOES NOT ELIMINATE BIAS IF PROXY VARIABLES REMAIN IN THE DATASET
- ❑ LATENT VARIABLES REQUIRE ADVANCED TECHNIQUES FOR DETECTION AND MITIGATION, ENSURING A FAIRER MODEL OVERALL

SIMPSON'S PARADOX

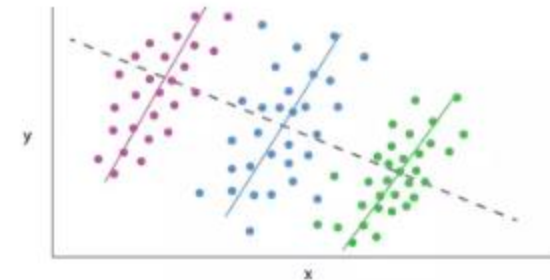
KEY DEFINITIONS

- ❑ SIMPSON'S PARADOX OCCURS WHEN A TREND OBSERVED IN AGGREGATED DATA REVERSES OR DISAPPEARS WHEN THE DATA IS DIVIDED INTO SUBGROUPS.
- ❑ RELYING SOLELY ON AGGREGATED DATA CAN LEAD TO MISLEADING CONCLUSIONS ABOUT FAIRNESS.

IMPLICATIONS FOR FAIRNESS IN ML

- ❑ OMITTING PROTECTED ATTRIBUTES WITHOUT SUBGROUP ANALYSIS MAY HIDE SIGNIFICANT DISPARITIES ACROSS DIFFERENT GROUPS
- ❑ A STRATIFIED APPROACH IS CRUCIAL TO IDENTIFY AND ADDRESS BIAS, ENSURING THE MODEL PERFORMS EQUITABLY FOR ALL GROUPS

Simpson's Paradox

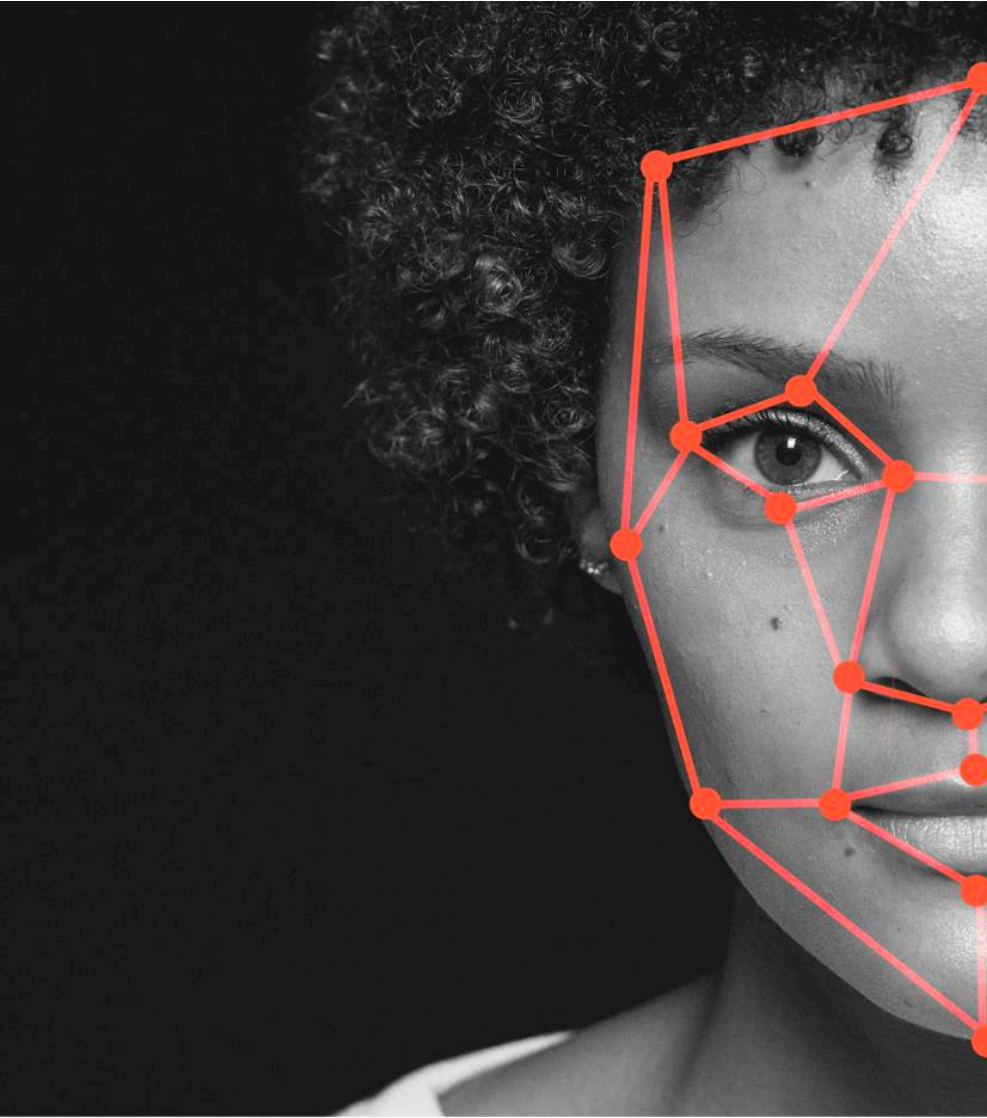


MEASURING FAIRNESS

MAIN MEASURES

- ❑ FAIRNESS THROUGH UNAWARENESS
- ❑ GROUP FAIRNESS
- ❑ CALIBRATION
- ❑ ERROR RATE BALANCE
- ❑ REPRESENTATIONAL FAIRNESS
- ❑ COUNTERFACTUAL FAIRNESS
- ❑ INDIVIDUAL FAIRNESS

<https://fairmlbook.org/>



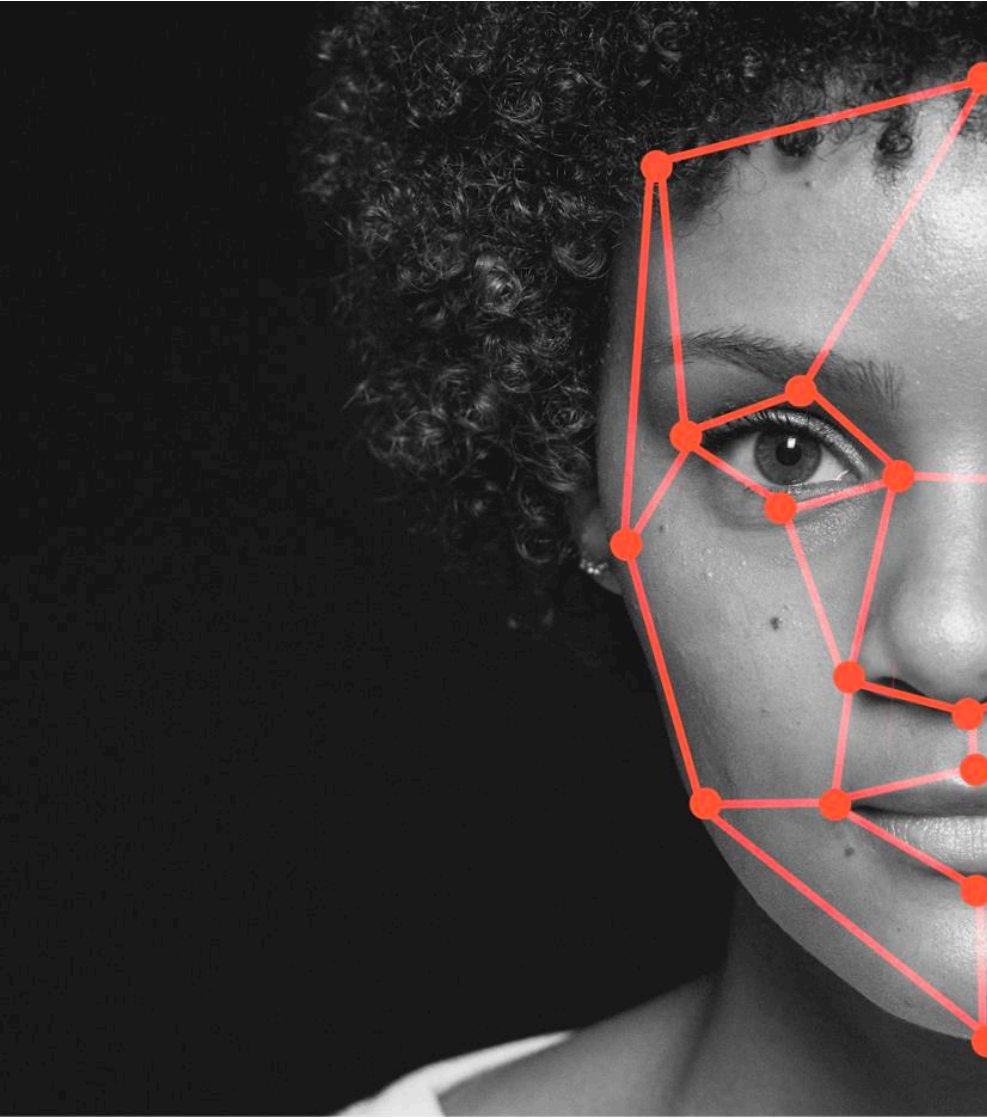
EVALUATING FOR BIAS

X FEATURES OF AN INDIVIDUAL

A SENSITIVE ATTRIBUTE (RACE, GENDER, ...)

$\hat{Y}=C(X,A)$ CLASSIFIER MAPPING X AND A TO SOME PREDICTION

Y ACTUAL OUTCOME



EVALUATING FOR BIAS

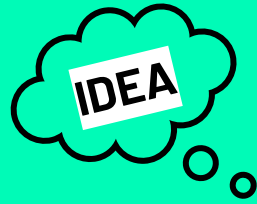
X INCORPORATES ALL SORTS OF MEASUREMENT BIASES

A OFTEN NOT EVEN KNOWN, ILL-DEFINED, MISREPORTED, INFERRED

C OFTEN NOT WELL DEFINED

$\hat{Y}$ OFTEN POOR PROXY OF ACTUAL VARIABLE OF INTEREST

FAIRNESS THROUGH UNAWARENESS



DON'T RECORD PROTECTED ATTRIBUTES, AND DON'T USE THEM IN YOUR ALGORITHM

- ❑ Predict risk Y from features X and group A using $P(\hat{Y} = Y | X)$ instead of $P(\hat{Y} = Y | X, A)$

PROS

GUARANTEED TO NOT BE MAKING A JUDGEMENT ON PROTECTED ATTRIBUTE

CONS

OTHER PROXIES MAY STILL BE INCLUDED IN A "RACE-BLIND" SETTING, E.G. ZIP CODE OR CONDITIONS

GROUP FAIRNESS



REQUIRE PREDICTION RATE BE THE SAME ACROSS PROTECTED GROUPS (E.G. "20% OF THE RESOURCES SHOULD GO TO THE GROUP THAT HAS 20% OF POPULATION")

- ❑ Predict risk Y from features X and group A such that $P(\hat{Y}=1|A=a) = P(A=a)$ for all a

PROS

LITERALLY TREATS EACH GROUP EQUALLY

CONS

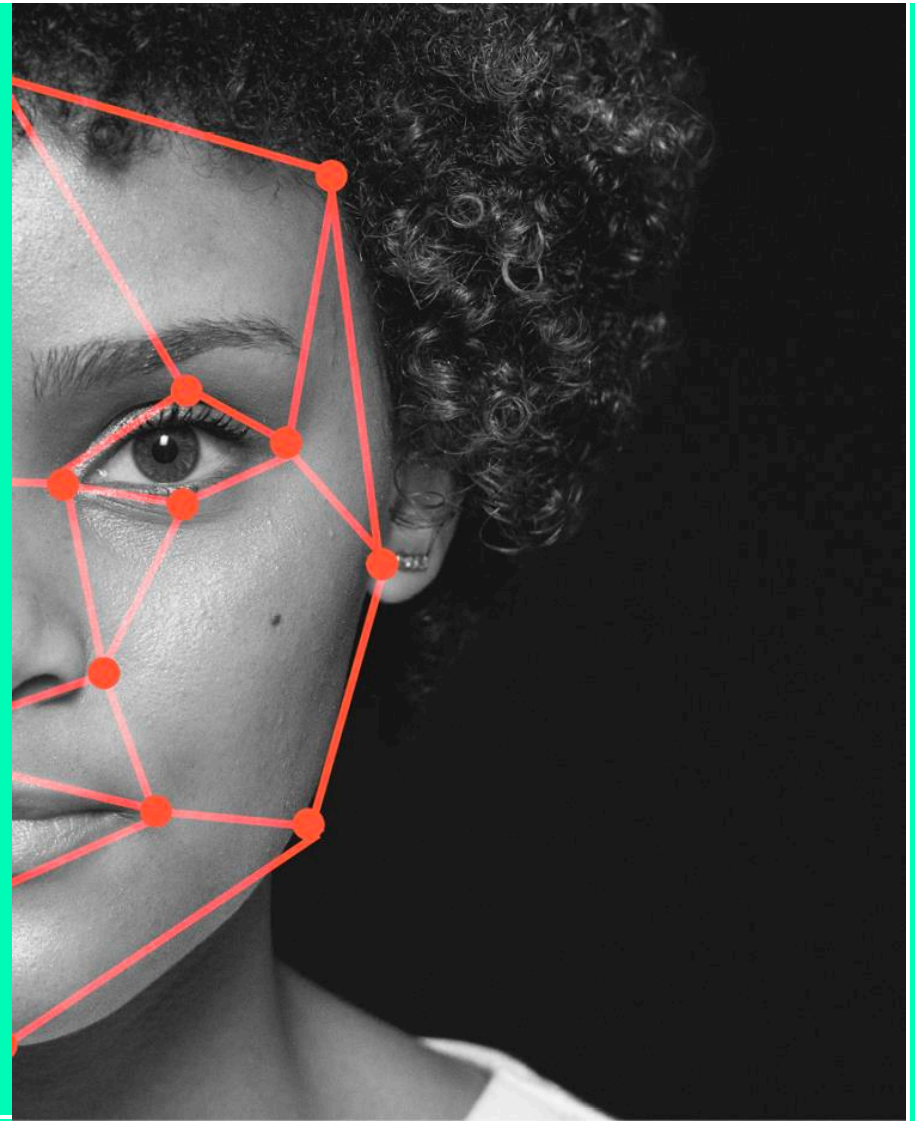
- ❑ **TOO STRONG:** GROUPS MIGHT HAVE DIFFERENT BASE RATES. THEN, EVEN A PERFECT CLASSIFIER WOULDN'T QUALIFY AS "FAIR"
- ❑ **TOO WEAK:** DOESN'T CONTROL ERROR RATE. COULD BE PERFECTLY BIASED (CORRECT FOR $A=0$ AND WRONG FOR $A=1$) AND STILL SATISFY.

DEMOGRAPHIC PARITY

ASSUME $\hat{Y}$ AND A ARE BINARY 0/1-VARIABLES

CLASSIFIER C SATISFIES DEMOGRAPHIC PARITY IF
 $\mathbb{P} \{ \hat{Y}=1 | A=1 \} = \mathbb{P} \{ \hat{Y}=1 | A=0 \}$

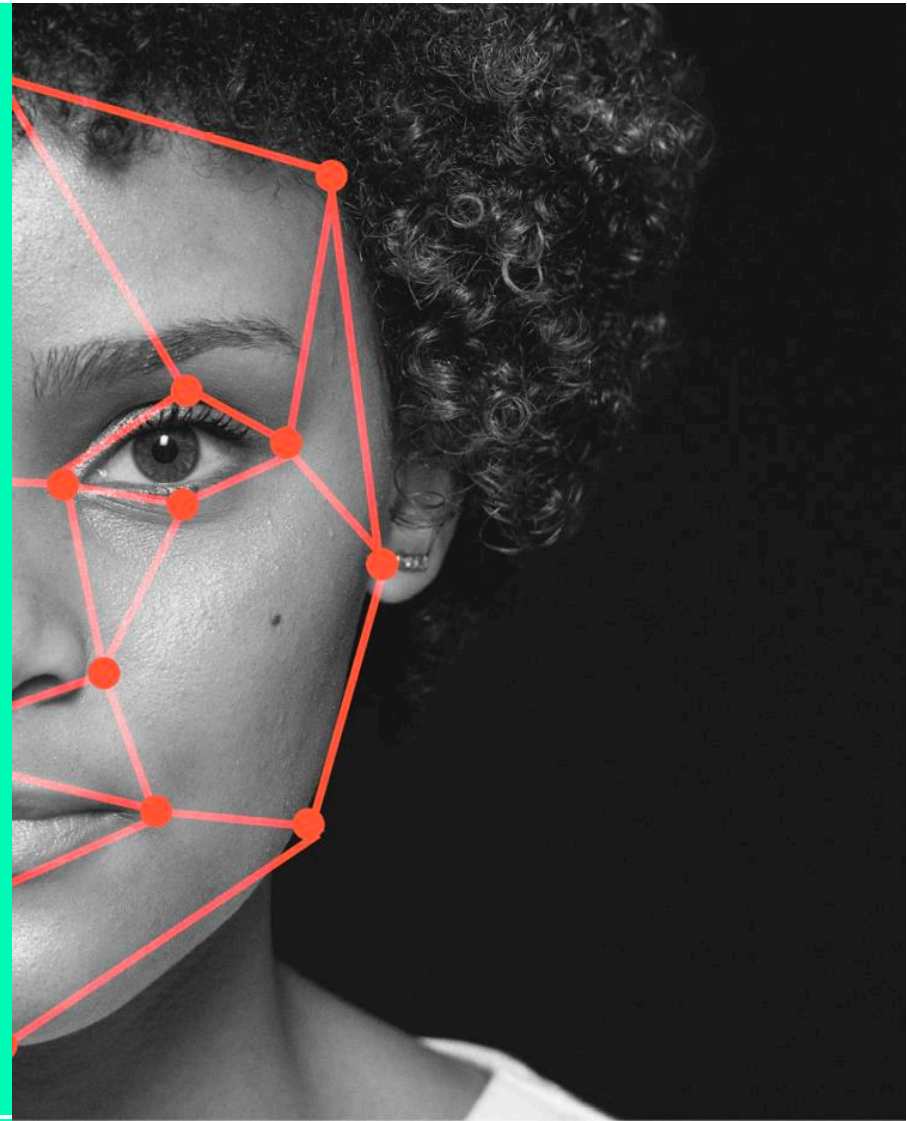
SAME PROS & CONS AS GROUP FAIRNESS



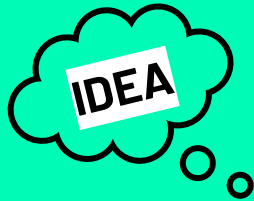
PRECISION PARITY

ASSUME $\hat{Y}, Y$ AND A ARE BINARY 0/1-VARIABLES

CLASSIFIER C SATISFIES ACCURACY PARITY IF
 $\mathbb{P}\{Y=1|\hat{Y}=1, A=1\} = \mathbb{P}\{Y=1|\hat{Y}=1, A=0\}$

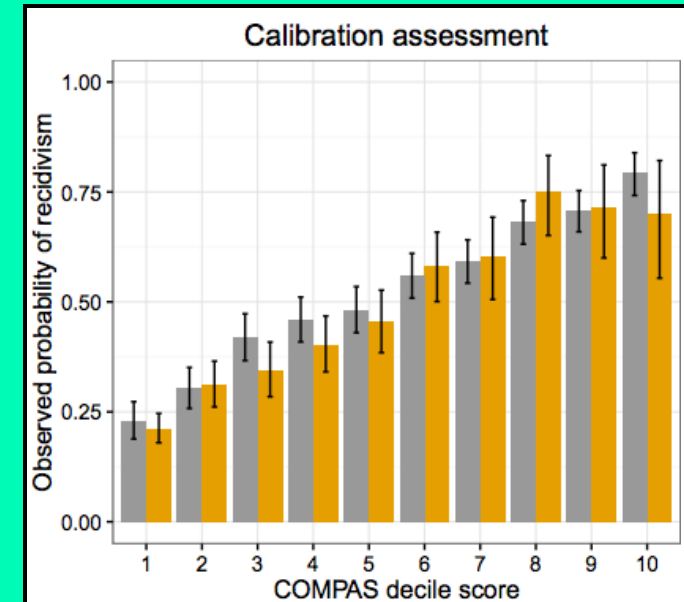


CALIBRATION



SAME POSITIVE PREDICTIVE VALUE ACROSS GROUPS

- Predict Y from features X and group A with score S:
 $P(Y = 1 | S = s, A = 1) = P(Y = 1 | S = s, A = 0)$



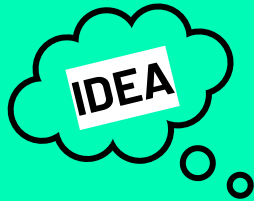
PROS

EQUALLY RIGHT ACROSS GROUPS

CONS

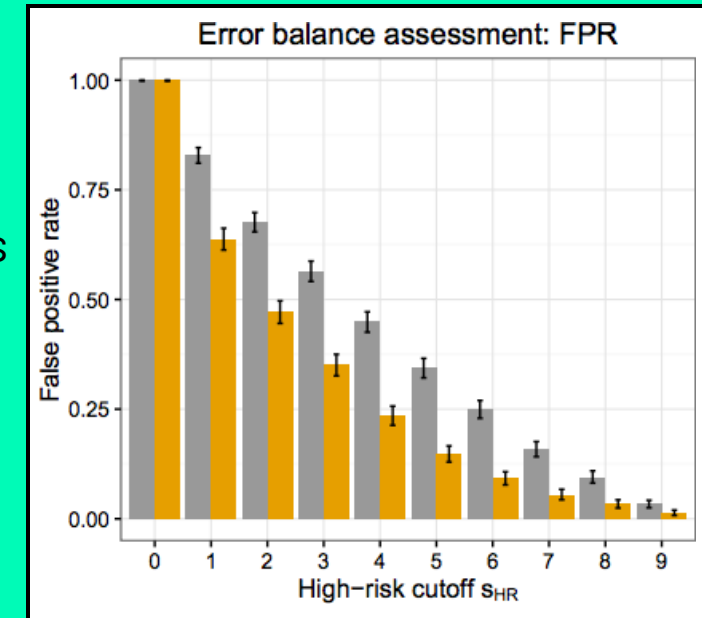
NOT COMPATIBLE WITH ERROR RATE
BALANCE (NEXT SLIDE)

ERROR RATE BALANCE



Equal false positive rates (FPR) across groups

$$\square P(\hat{Y}=1 \mid Y=0, A=1) = P(\hat{Y}=1 \mid Y=0, A=0)$$



PROS

EQUALLY WRONG ACROSS GROUPS

CONS

INCOMPATIBLE WITH CALIBRATION AND FALSE NEGATIVE RATES (FNR), COULD DILUTE WITH EASY CASES

Inherent Trade-Offs in the Fair Determination of Risk Scores

Jon Kleinberg *

Sendhil Mullainathan [†]

Manish Raghavan [‡]

Abstract

Recent discussion in the public sphere about algorithmic classification has involved tension between competing notions of what it means for a probabilistic classification to be fair to different groups. We formalize three fairness conditions that lie at the heart of these debates, and we prove that except in highly constrained special cases, there is no method that can satisfy these three conditions simultaneously. Moreover, even satisfying all three conditions approximately requires that the data lie in an approximate version of one of the constrained special cases identified by our theorem. These results suggest some of the ways in which key notions of fairness are incompatible with each other, and hence provide a framework for thinking about the trade-offs between them.

Inherent Trade-Offs in the Fair Determination of Risk Scores

Jon Kleinberg *

Sendhil Mullainathan †

Manish Raghavan ‡

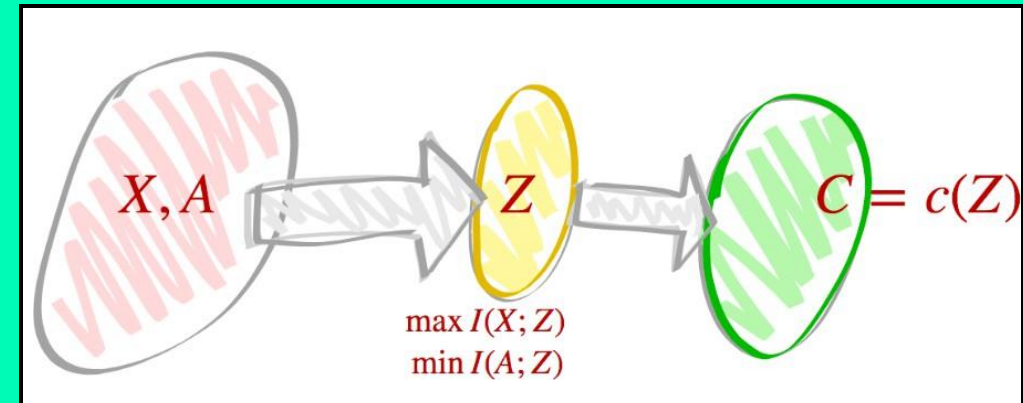
“WE PROVE THAT EXCEPT IN HIGHLY CONSTRAINED
SPECIAL CASES, THERE IS NO METHOD THAT SATISFIES
THESE THREE [FAIRNESS] CONDITIONS
SIMULTANEOUSLY”

of the ways in which key notions of fairness are incompatible with each other, and hence provide a framework for thinking about the trade-offs between them.

REPRESENTATIONAL FAIRNESS



LEARN LATENT
REPRESENTATION Z TO
MINIMIZE GROUP
INFORMATION



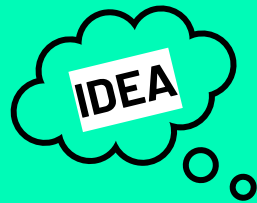
PROS

REDUCE INFORMATION GIVEN TO
MODEL BUT STILL KEEP IMPORTANT
INFO

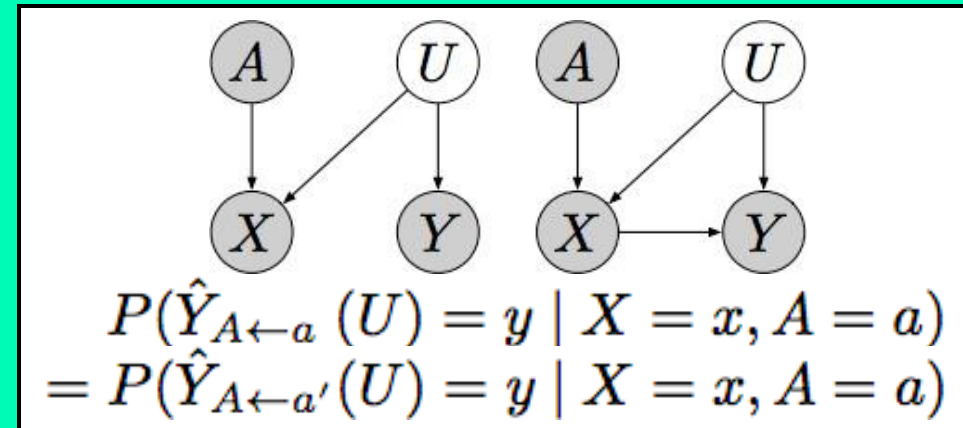
CONS

TRADE-OFF BETWEEN ACCURACY
AND FAIRNESS

COUNTERFACTUAL FAIRNESS



GROUP A SHOULD NOT
CAUSE PREDICTION $\hat{Y}$



PROS

CAN MODEL EXPLICIT CONNECTIONS
BETWEEN VARIABLES

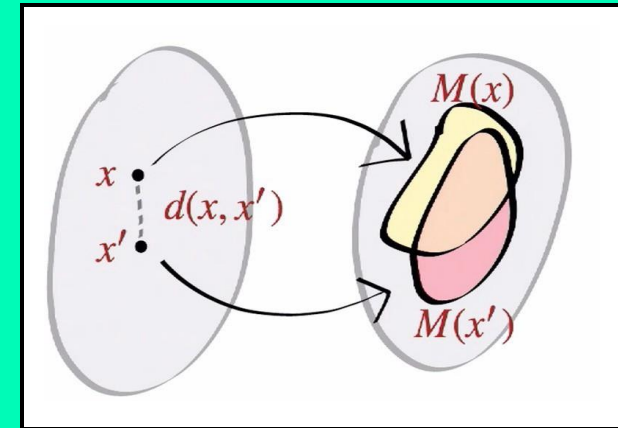
CONS

- ❑ GRAPH MODEL MAY NOT ACTUALLY REPRESENT WORLD
- ❑ INFERENCE ASSUMES OBSERVED CONFOUNDERS

INDIVIDUAL FAIRNESS



SIMILAR INDIVIDUALS
SHOULD BE TREATED
SIMILARLY



PROS

CAN MODEL HETEROGENEITY WITHIN
EACH GROUP

CONS

NOTION OF "SIMILAR" IS HARD TO
DEFINE MATHEMATICALLY,
ESPECIALLY IN HIGH DIMENSIONS

MAIN MEASURES

☒ ~~FAIRNESS THROUGH UNAWARENESS~~

NOT USEFUL

☐ GROUP FAIRNESS

☐ CALIBRATION

MORE STANDARD

☐ ERROR RATE BALANCE

☐ REPRESENTATIONAL FAIRNESS

☐ COUNTERFACTUAL FAIRNESS

MORE EXPERIMENTAL

☐ INDIVIDUAL FAIRNESS

**IF YOU WERE THE DEVELOPERS
OF COMPAS AND NOTICED THOSE
BIASES (E.G., HIGH FALSE
POSITIVE RATES FOR BLACK
INDIVIDUALS), WHAT WOULD
YOU DO TO ADDRESS THEM?**

OUTLINE

1

INTRODUCTION TO BIAS AND FAIRNESS IN AI



2

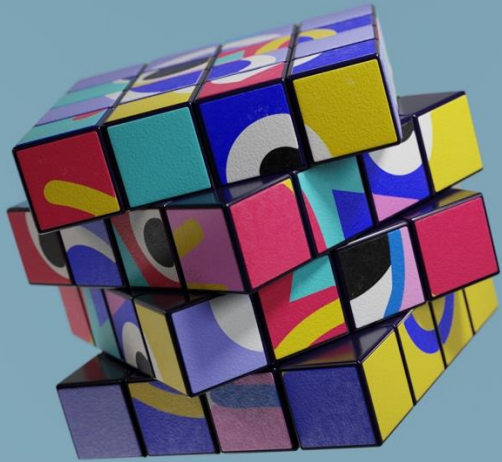
FAIRNESS MEASURES



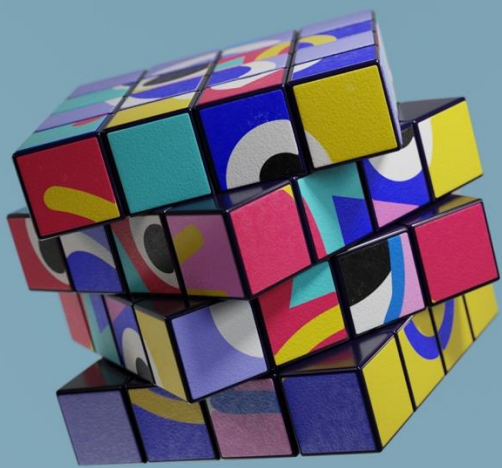
3

BIAS MITIGATION





BIAS MITIGATION



Pre-processing





Pre-processing



In-processing



Pre-processing



In-processing



Post-processing



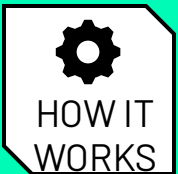
RE-SAMPLING



BALANCE THE DATASET BY DUPLICATING UNDERREPRESENTED EXAMPLES OR REMOVING OVERREPRESENTED EXAMPLES



ENSURES BALANCED REPRESENTATION OF DIFFERENT GROUPS



DUPLICATE RECORDS OF UNDERREPRESENTED GROUP CASES (E.G., NON-RECIDIVIST BLACK DEFENDANTS) OR REMOVE EXCESSIVE RECORDS FROM OVERREPRESENTED GROUPS (E.G., RECIDIVIST BLACK DEFENDANTS)

PROS

- ❑ SIMPLE AND EFFECTIVE AT REDUCING GROUP DISPARITIES

CONS

- ❑ CAN LEAD TO OVERFITTING DUE TO DUPLICATION
- ❑ POTENTIALLY REDUCES DATA DIVERSITY BY REMOVING EXAMPLES

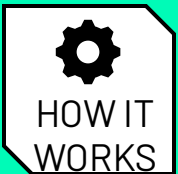
RE-WEIGHTING



ASSIGN HIGHER TRAINING WEIGHTS TO UNDERREPRESENTED EXAMPLES



GUIDES THE MODEL TO PAY MORE ATTENTION TO UNDERREPRESENTED CASES DURING LEARNING



ASSIGN HIGHER WEIGHTS TO CASES OF WHITE RECIDIVISM OR BLACK NON-RECIDIVISM TO CORRECT SYSTEMIC BIAS IN THE DATASET

PROS

- ❑ PRESERVES DATA WITHOUT REMOVAL OR DUPLICATION.
- ❑ SIMPLE TO IMPLEMENT

CONS

- ❑ SENSITIVE TO CHOICE OF WEIGHTS; POOR CALIBRATION CAN INTRODUCE NEW BIASES

**HOW COULD WE
IMPLEMENT
RE-WEIGHTING?**

RE-WEIGHTING

Step 1: Calculate overall prevalence of the positive class
 $\text{prevalence_p} = P(Y=1)$ *# global percentage of positives*



How frequent is the positive class in the entire dataset?

Example: If 300 out of 1000 instances have $Y=1$, then $\text{prevalence_p} = 0.3$. This helps us spot underrepresentation

RE-WEIGHTING

Step 1: Calculate overall prevalence of the positive class

`prevalence_p = P(Y=1)` *#global percentage of positives*

Step 2: For each instance in the dataset, assign a weight

For each instance *i* in the dataset:

If `i.group == "Black"` and `i.label == 1`:

`weight[i] = 1 / prevalence_p` *# assign higher weight: if positives are rare, their weight increases*

📌 If the instance belongs to an underrepresented group and has a positive label:

- Assign it a higher weight, inversely proportional to prevalence
- If positives are rare (`prevalence_p` is low), the weight increases

💡 Intuition: if only a few positives come from a group, we need to make them “count more” during training

RE-WEIGHTING

Step 1: Calculate overall prevalence of the positive class
prevalence_p = $P(Y=1)$ *#global percentage of positives*

Step 2: For each instance in the dataset, assign a weight
For each instance i in the dataset:

If $i.\text{group} == \text{"Black"}$ and $i.\text{label} == 1$:

$\text{weight}[i] = 1 / \text{prevalence_p}$ *# assign higher weight: if
positives are rare, their weight
increases*

Else:

$\text{weight}[i] = 1$

→ Other instances (e.g., majority group or negatives) are not reweighted

RE-WEIGHTING

Step 1: Calculate overall prevalence of the positive class

`prevalence_p = P(Y=1)` *#global percentage of positives*

Step 2: For each instance in the dataset, assign a weight

For each instance *i* in the dataset:

 If `i.group == "Black"` and `i.label == 1`:

`weight[i] = 1 / prevalence_p` *# assign higher weight: if
positives are rare, their weight
increases*

 Else:

`weight[i] = 1`

Step 3: Use these weights during model training

For example, in the model's loss function

`loss = Sum(weight[i] * model_loss(i))`



The model will update its parameters by giving more importance to errors on underrepresented groups. This way, those groups influence learning more, even if they are fewer in number

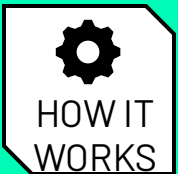
FEATURE TRANSFORMATION



TO REDUCE OR ELIMINATE CORRELATIONS BETWEEN A SENSITIVE ATTRIBUTE (E.G., RACE) AND OTHER VARIABLES (FEATURES) IN THE DATASET.



EVEN IF YOU DIRECTLY REMOVE THE SENSITIVE ATTRIBUTE, OTHER INDIRECTLY CORRELATED FEATURES MIGHT STILL ALLOW THE MODEL TO INFER THE SENSITIVE INFORMATION →
SURROGATE CORRELATION



TWO MODELS ARE TRAINED ADVERSARIALLY: ONE LEARNS TO CREATE DATA REPRESENTATIONS THAT RETAIN RELEVANT PREDICTIVE INFORMATION WHILE REMOVING SIGNALS RELATED TO A SENSITIVE ATTRIBUTE; THE OTHER TRIES TO PREDICT THAT SENSITIVE ATTRIBUTE. THE GOAL IS TO PRODUCE REPRESENTATIONS THAT ARE INFORMATIVE FOR THE MAIN TASK BUT UNINFORMATIVE ABOUT THE SENSITIVE VARIABLE

PROS

- ❑ REDUCES INDIRECT DISCRIMINATORY ASSOCIATIONS EFFECTIVELY
- ❑ RETAINS USEFUL PREDICTIVE INFORMATION

CONS

- ❑ COMPUTATIONALLY INTENSIVE AND COMPLEX
- ❑ POTENTIAL INFORMATION LOSS IF NOT CALIBRATED PROPERLY

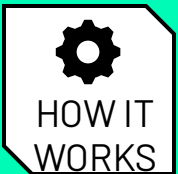
REGULARIZATION/CONSTRAINT



INCORPORATE FAIRNESS CONSTRAINTS INTO THE MODEL'S LOSS FUNCTION



ENSURES FAIRNESS OBJECTIVES ARE MET DIRECTLY DURING MODEL TRAINING



PENALIZE THE MODEL IF PREDICTIONS DIFFER SIGNIFICANTLY BETWEEN SENSITIVE GROUPS (E.G., RACIAL GROUPS)

PROS

- ❑ DIRECTLY OPTIMIZES FAIRNESS DURING TRAINING

CONS

- ❑ MAY REDUCE OVERALL PREDICTIVE PERFORMANCE
- ❑ REQUIRES CAREFUL TUNING OF CONSTRAINT STRENGTH

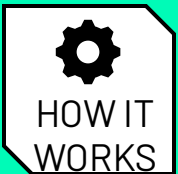
LEARNING FAIR REPRESENTATIONS



LEARN REPRESENTATIONS OF DATA EXPLICITLY FREE OF BIAS RELATED TO SENSITIVE ATTRIBUTES



PRODUCES INHERENTLY FAIRER PREDICTIONS BY DESIGN



LEARN EMBEDDINGS OR FEATURE TRANSFORMATIONS WHERE RACE CANNOT BE PREDICTED FROM TRANSFORMED DATA

PROS

- ❑ ADDRESSES BIAS PROACTIVELY AT REPRESENTATION LEVEL

CONS

- ❑ MORE COMPLEX IMPLEMENTATION
- ❑ POSSIBLE LOSS OF USEFUL PREDICTIVE INFORMATION

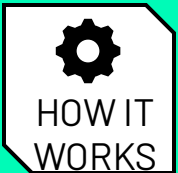
ADVERSARIAL DEBIASING



TRAIN MODELS USING AN ADVERSARIAL SETUP TO ENSURE SENSITIVE ATTRIBUTES CANNOT BE INFERRED FROM LEARNED REPRESENTATIONS



ACTIVELY ENCOURAGES UNBIASED MODEL REPRESENTATIONS



AN ADVERSARIAL CLASSIFIER TRIES TO PREDICT RACE FROM LEARNED REPRESENTATIONS, AND THE MAIN MODEL LEARNS REPRESENTATIONS THAT FOOL THE ADVERSARY

PROS

- ❑ HIGHLY EFFECTIVE AT REDUCING INDIRECT BIAS

CONS

- ❑ COMPUTATIONALLY EXPENSIVE
- ❑ COMPLEX TO IMPLEMENT AND OPTIMIZE

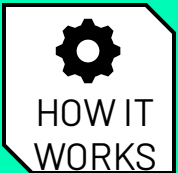
ADJUSTING DECISION THRESHOLD



ADJUST DECISION THRESHOLDS FOR DIFFERENT SENSITIVE GROUPS



ENSURES EQUAL OUTCOMES ACROSS GROUPS



USE A HIGHER THRESHOLD FOR "HIGH-RISK" PREDICTIONS AMONG HISTORICALLY OVERREPRESENTED GROUPS

PROS

- ❑ EASY TO APPLY TO EXISTING MODELS

CONS

- ❑ ONLY ADJUSTS OUTCOMES SUPERFICIALLY WITHOUT ADDRESSING UNDERLYING BIAS

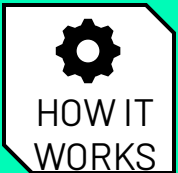
CALIBRATED POST-PROCESSING



ADJUST PREDICTED PROBABILITIES TO ACHIEVE FAIRNESS CONSTRAINTS (E.G., EQUAL POSITIVE RATES)



MAINTAINS PREDICTIVE QUALITY WHILE MEETING FAIRNESS CONSTRAINTS



ADJUST PREDICTED PROBABILITIES PER GROUP TO ACHIEVE A SPECIFIC METRIC

PROS

- ❑ DIRECTLY ACHIEVES DESIRED FAIRNESS METRICS

CONS

- ❑ REQUIRES CAREFUL CALIBRATION
- ❑ CAN DEGRADE PREDICTION QUALITY IF MISAPPLIED

TOOLKIT AND LIBRARIES

IBM AI Fairness 360 (AIF360)

Language: Python

Focus: Fairness metrics and bias mitigation techniques

Use cases: Classification, structured data, pre-processing, in-processing, post-processing

Website: <https://aif360.res.ibm.com/>

GitHub: [AIF360 GitHub](#)

Includes over 70 fairness metrics and 10 bias mitigation algorithms

Fairlearn

Language: Python

Focus: Fairness-aware model assessment and selection

Integrates with: scikit-learn, Azure ML

GitHub: [Fairlearn GitHub](#)

Supports disparity measurement and mitigation strategies such as equalized odds and demographic parity

**CONSIDERING TWO DIFFERENT
SCENARIOS, COMPAS AND HIRING
ALGORITHM, WHICH BIAS
MITIGATION STRATEGIES WOULD
YOU APPLY, AND WHY?**



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THANK YOU!